AthenaStudies

Intelligent Systems

Free Course | Mara Spadon | 2023

Welcome!

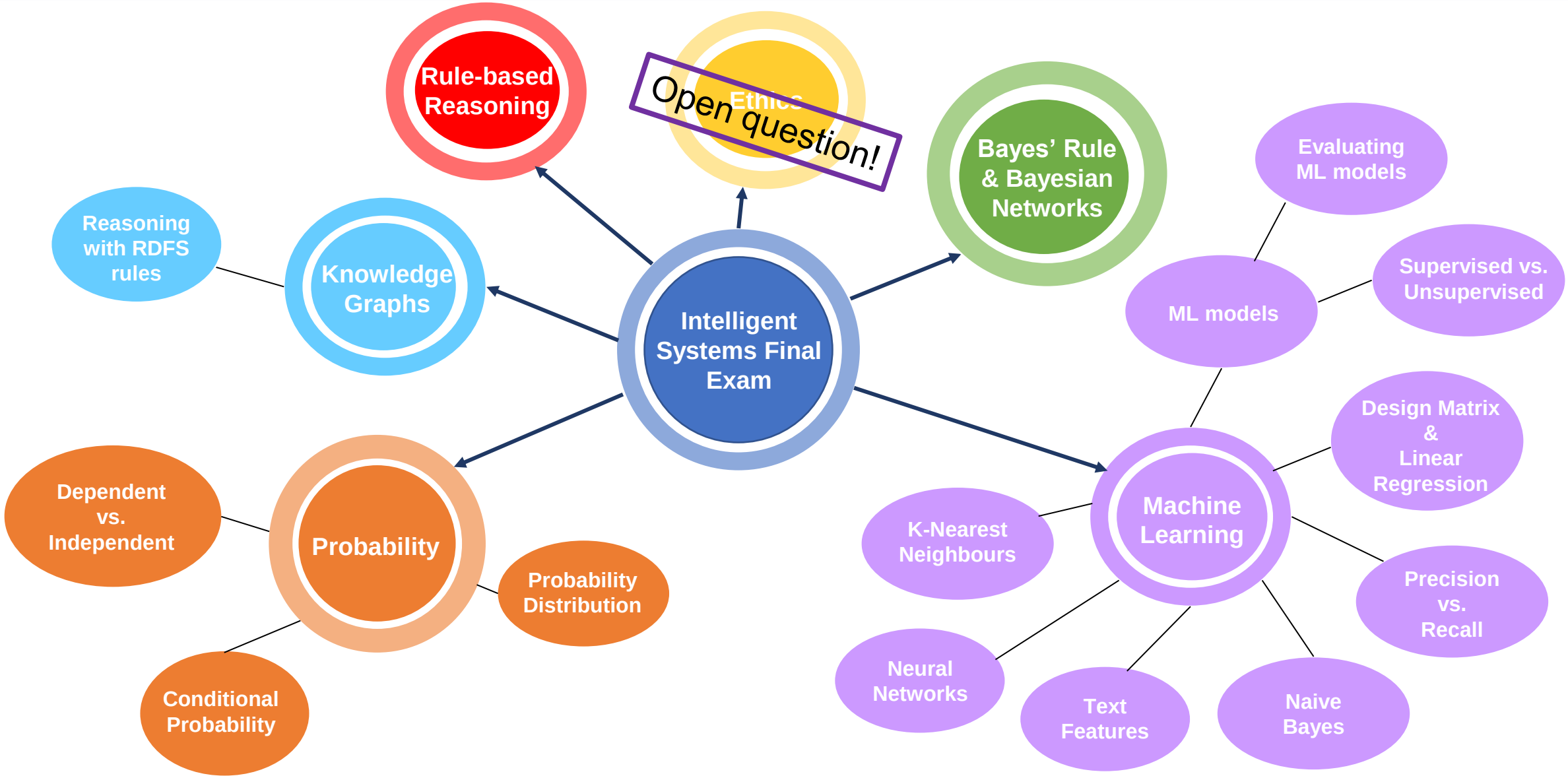
Mara Spadon

About me

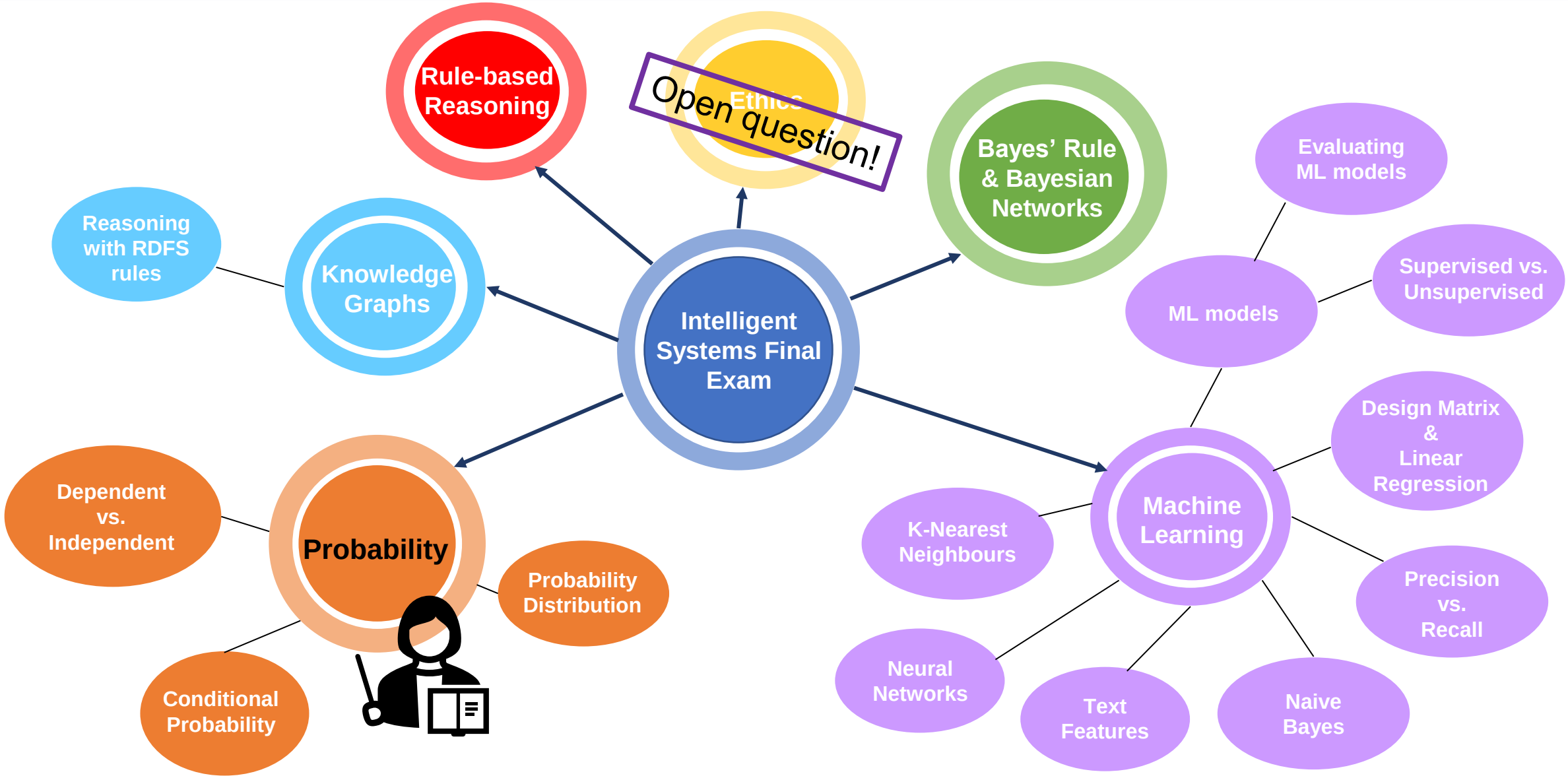
- Bachelor's Degree AI
- TA Intelligent Systems & Athena Studies
- Past experience with teaching this course

WhatsApp

Exam analysis



Exam analysis



Pro-active participation

- Camera's on (optional)
- Raise your hand to ask questions / Ask questions in the chat!

- Be interactive!
- Take notes!
- It is okay to make mistakes
- Slides will be sent afterwards

Structure

- Structure based on past exams
- Learn the most by asking questions
- Active learning with old exam questions is the basis of our trainings

Planning

Planning for today
Introduction
Probability Theory
Break
Exam questions

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Probability Theory

Sample Space

Sample space: $\Omega = \{...\}$

It is the set of single outcomes of our experiment

die roll $\Omega = \{1, 2, 3, 4, 5, 6\}$

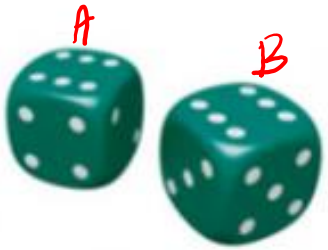
Can it also include pairs? $\Rightarrow$ yes!

2 dice $\Omega = \{(1;4), (5;3), \dots (6;6)\}$

Sample space examples

$\Omega = \{\text{heads, tails}\}$

$\Omega = \{1, 2, 3, 4, 5, 6\}$



$\Omega = \{(\overset{A}{1};\overset{B}{1}), (\overset{A}{1};\overset{B}{2}), (2;1), (2;2), \ldots, (6;6)\}$

Events

Subsets of the sample space, all possible events

$\Omega = \{1, 2, 3, 4, 5, 6\}$

What are all possible events of a die roll?

E = {...}

Roll an even n $E = \{2, 4, 6\}$

Roll a 3 $E = \{3\}$

Roll 1 or 3 $E = \{1, 3\}$

Roll 1 and 3 $E = \{\}$

Events and Probability

$$E = \{\{\}, \{1\}, \{2\}, \{3\}, \ldots, \{1, 2\}, \ldots, \{1, 2, 3, 4, 5, 6\}\}$$

Using events to calculate probability:

P({1,2,3})	*probability that outcome < 4*
P({2,4,6})	*probability that outcome is an even number*
P({6})	*probability that outcome = 6*

Probability Distribution

The probability distribution gives a value for all possible assignments, denoted with **P**

→ bold

When asked for **P**(…), give your answer in a vector of probability values:

<…= True, …= False>

<0.4, 0.6> → sum = 1

The value of the probability vector should always lead to a sum of 1

A	B	C	P(A, B, C)
false	false	false	0.05
false	false	true	0.10
false	true	false	0.03
false	true	true	0.25
true	false	false	0.15
true	false	true	0.02
true	true	false	0.07
true	true	true	0.33

→ sum = 1

Conditional Probability

"given"

$P(A|B)$

"The probability of A given B"

Conditional Probability Formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Probability of *A* given *B*

Probability of *A* and *B*

Probability of *B*

Conditional Probability

$\Omega = \{1, 2, 3, 4, 5, 6\}$

What is P({3}| uneven)?

$$P(A|B) = \frac{P(A,B)}{P(B)}$$

$P(\{3\}, \text{uneven}) = 1/6$

$P(\text{uneven}) = 3/6 = 1/2$

$$P(\{3\} | \text{uneven}) = \frac{1/6}{1/2} = 1/3$$

Inference by enumeration

P(Toothache, Catch, Cavity)

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

For any proposition ϕ, sum the atomic events where it is true: $P(\phi) = \Sigma_{\omega:\omega \models \phi} P(\omega)$

What does this mean for P(cavity)? → prob that cavity = true

$.108 + .012 + .072 + .008 = 0.2$

Recap Question: Probability Distribution

What is P(B = true)?

$0.03 + 0.25 + 0.07 + 0.33$
$= 0.68$

What is P(A & B = true)?

$0.07 + 0.33 = 0.4$

What is P(B v C = true)?

A	B	C	P(A, B, C)
false	false	false	0.05
false	false	true	0.10
false	true	false	0.03
false	true	true	0.25
true	false	false	0.15
true	false	true	0.02
true	true	false	0.07
true	true	true	0.33

sum 1

① $0.10 + 0.03 + 0.25 + 0.02 + 0.07 + 0.33 = 0.8$

② $\boxed{\begin{matrix} B=F \\ C=F \end{matrix}}$ $0.2 \Rightarrow 1 - 0.2 = 0.8$ (same answer)

Recap Question: Probability

	toothache		¬ toothache	
	catch	¬ catch	catch	¬ catch
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

$$P(A|B) = \frac{P(A,B)}{P(B)}$$

Given this full joint probability distribution, calculate the following:

1. P(-toothache | cavity)

A B

$P(A,B) = 0.072 + 0.008 = 0.08$

$P(B) = 0.2$

$\Rightarrow \frac{0.08}{0.2} = 0.4$

2. **P**(Cavity)

$\langle cavity, \neg cavity \rangle$

sum to 1

$P(cavity) = 0.2$

$P(\neg cavity) = 0.8$ $(1-0.2)$

$\Rightarrow \langle 0.2, 0.8 \rangle$

Bold P / Capital letter

(bold P)

$\mathbb{P}$ = give a probability vector

normal capital

P = probability

$\langle \ldots = \text{true}, \ldots = \text{false} \rangle$

$\Rightarrow \langle \text{cavity}, \neg\text{cavity} \rangle$

sum

1

Capital letter = give the value and ¬value

$\mathbb{P}(\text{Cavity}) = ?$ $\quad P(\text{cavity})$, $P(\neg\text{cavity})$

$\mathbb{P}(\text{Toothache} \mid \text{catch}) = P(\overset{A}{\text{tooth}} \mid \text{catch})$, $P(\underset{B}{\neg\text{tooth}} \mid \text{catch})$

$\langle A, B \rangle \rightarrow$ sum to 1

(In)dependent events

$$P(A|B) = P(A)$$

B has no effect on A.

A and B are independent

$$P(A|B) \neq P(A)$$

B has effect on A

A and B are dependent

are these two events (in)depen. ?

1. calculate $P(A|B)$
2. Calculate $P(A)$
3. check if $P(A|B) = P(A)$ or not
4. determines (in)dependence

Recap Question: (in)dependence

You throw two fair dice, one green and one red, and observe the numbers uppermost. Decide whether the following pair of events are (in)dependent.

- A: the sum is 5
- B: the red dice shows a 2

$P(A|B) \neq, = P(A)$

- $P(A) \Rightarrow P(\text{sum is } 5) = 4/36 = 1/9$

$r = 1, 2, 3, 4, 5, 6$
$g = 1, 2, 3, 4, 5, 6$

$6^2 = 36$

$1+4,\ 2+3,\ 4+1,\ 3+2 \Rightarrow 4$ events

- $P(A|B) = P(\text{sum } 5 \mid r = 2) = 1/6$

$r = 2$ — 1 event
$g = 1, 2, 3, 4, 5, 6$
6

- $1/6 \neq 1/9$, dependent events

Planning

Planning for today
Introduction ✓
Probability Theory ✓
Break
Exam questions

AthenaStudies

Break:
5-10 minutes

17:00
- 17:30

Planning

Planning for today
Introduction ✓
Probability Theory ✓
Break ✓
Exam questions

Exam Questions

Exam question: Probability

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

Given the full joint distribution as above, calculate the following:

P(Toothache | cavity)

$\langle \ldots, \ldots \rangle$

$$P(\text{tooth} \mid \text{cavity}) = (t \cap c)/c = \frac{0.108 + 0.12}{0.2} = 0.6$$

$$P(\neg\text{tooth} \mid \text{cavity}) = (\neg t \cap c)/c = \frac{0.72 + 0.008}{0.2} = 0.4$$

$1 - 0.6$

$\Downarrow$

$\langle 0.6, 0.4 \rangle$

$\downarrow$

check $= 1$

send me privately your answer!

Exam question: Probability

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

0.2

Given the full joint distribution as above, calculate the following:

P(Toothache | cavity)

P(cavity) = 0.2

P(toothache | cavity) = (0.108 + 0.12) / 0.2 = 0.6
P(-toothache | cavity) = (0.72 + 0.008) / 0.2 = 0.4

P(Toothache | cavity) = <0.6,0.4>

Exam question: Probability

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

Given the full joint distribution as above, calculate the following:

P(Cavity | toothache, catch)

Exam question: Probability

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

Given the full joint distribution as above, calculate the following:

P(Cavity | toothache, catch)

P(toothache, catch) = 0.416

P(cavity | t,c) = (.108+.012+.072)/0.416 = 0.4615
P(-cavity | t,c) = (.016+.064+.144)/0.416 = 0.5384

P(Cavity | toothache, catch) = <0.4615, 0.5384>

Exam question: Joint Probability Distribution

Question 7 – Multiple response – Question-ID: 217964 (3 points)

Consider the following full joint probability distribution:

P	Q	R	P(P, Q, R)
true	true	true	0.26
true	true	false	0.33
true	false	true	0.04
true	false	false	0.14
false	true	true	0.03
false	true	false	0.08
false	false	true	0.11
false	false	false	0.01

Which statements are true?

A P(P = true v Q =true v R = true) = 1

B P(R = false) = 0.56 ✓

C P(Q=true|R-true) = P(R=true|Q=true)

D P(Q = true) = 0.26

E P(P = true & Q = true) = 0.59 ✓

F P(R = false) = 0.63

Exam question: Joint Probability Distribution

Question 7 – Multiple response – Question-ID: 217964 (3 points)

Consider the following full joint probability distribution:

P	Q	R	P(P, Q, R)
true	true	true	0.26
true	true	false	0.33
true	false	true	0.04
true	false	false	0.14
false	true	true	0.03
false	true	false	0.08
false	false	true	0.11
false	false	false	0.01

= 0.59

Which statements are true?

A P(P = true v Q =true v R = true) = 1 ⇒ 0.99

B P(R = false) = 0.56

C P(Q=true|R-true) = P(R=true|Q=true) → conditional prob. $\boxed{A|B \neq B|A}$ not given

D P(Q = true) = 0.26

E P(P = true & Q = true) = 0.59

F P(R = false) = 0.63

Exam question: Probability

Which of the following expressions is **not symmetric**?

A) Union probability: P(AUB) is not always equal to P(BUA)

B) Joint probability: P(A, B) is not always equal to P(B, A)

C) Conditional probability: P(A|B) is not always equal to P(B|A)

Exam question: Probability

Which of the following expressions is **not symmetric**?

A) Union probability: P(AUB) is not always equal to P(BUA)

B) Joint probability: P(A, B) is not always equal to P(B, A)

C) Conditional probability: P(A|B) is not always equal to P(B|A)

Evaluation form

Step 1

Scan the QR code
or check your email

Step 2

Answer the 5
questions

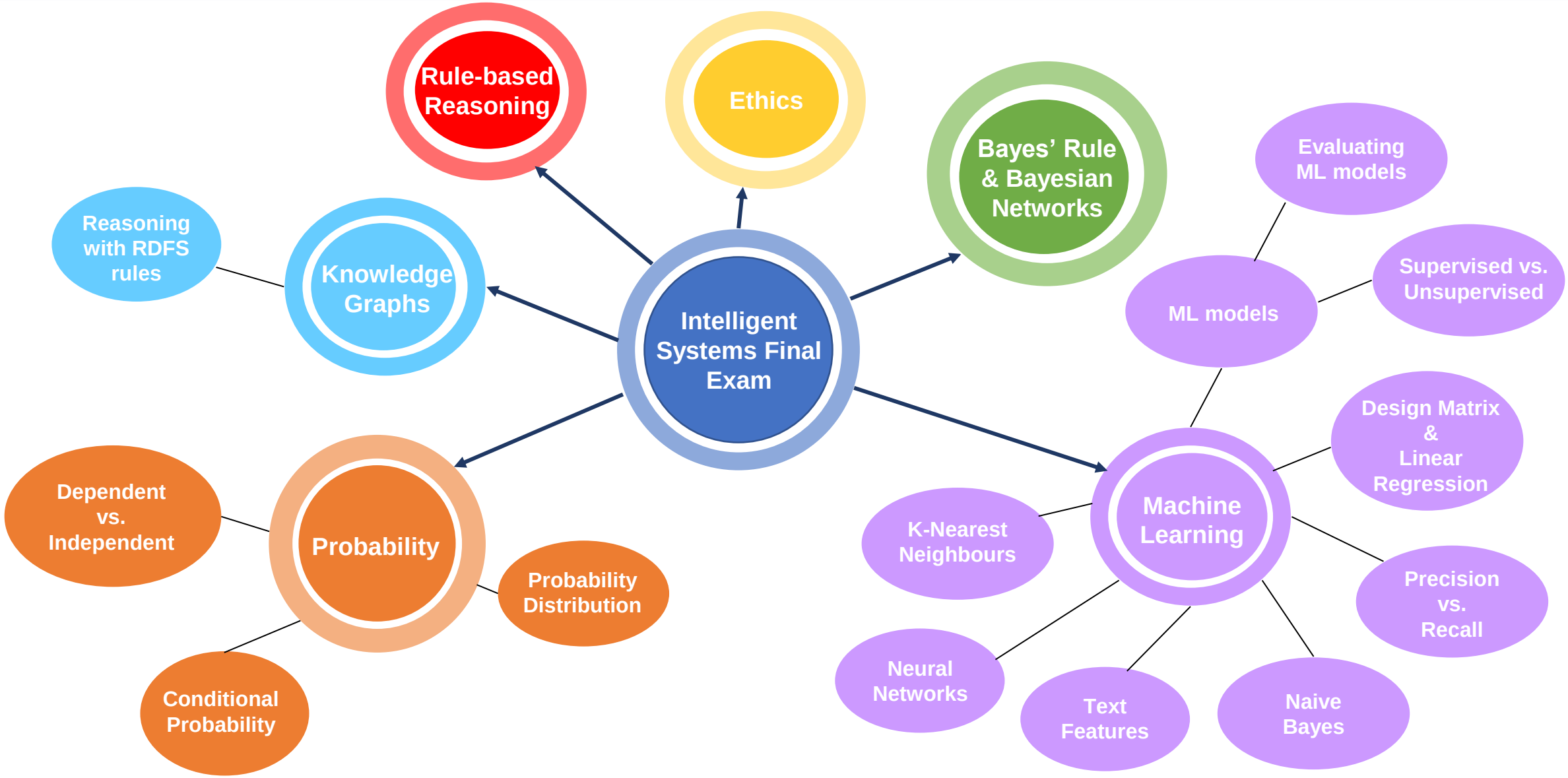
Step 3

Sit back & relax

Free vs full

	Freee AthenaStudies course	Full AthenaStudies exam training
Precisely explained which topics you can expect on the exam	✓	✓
All practice questions based on **past exams**	✓	✓
All exam material covered		✓
Small groups, room for **all your questions**		✓
24/7 WhatsApp aid		✓

Exam analysis



Full Course

About this course

Tutor: **Mara Spadon**

- 3rd-year **Artificial Intelligence** student (**GPA: 8**)
- Intelligent Systems **Teaching Assistant**
- Attained **8.5** for Intelligent Systems

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Mo 18 Dec	13:00 - 17:00

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In cart

Exam question: Joint Probability Distribution

(3 Points) Consider the following joint probability distribution on random variables A which takes the values 10, 20 and 30 and B which takes the values x, y and z. We will abbreviate p(A=10 & B = z) as p(10, z).

- p(10, x) = 1/6
- p(10, y) = 1/60
- p(10, z) = 1/10
- p(20, x) = 1/60
- p(20, y) = 1/12
- p(20, z) = 1/5
- p(30, x) = 7/20
- p(30, y) = 0/60
- p(30, z) = ?

sum 1

Which of the following statements are correct (multiple answers possible)? (4 correct)

- [ ] p(30, z) cannot be computed from the information given A
- [x] p(x) = 32/60 B ✓ $1/6 + 1/60 + 7/20 = 8/15$ ↳ $32/60$
- [x] **p(A & B) = p(B & A) for all values of A and B** C ✓
- [ ] **p(20 | y) = 5/60** D
- [x] **p(30, z) = 1/15** E 1 - everything given = $1/15$ ✓
- [x] **p(y | 30) = p(30 | y)** $0/60 = 0$ F ✓
- [ ] **p(X|Y) = P(Y|X) is never true for all values of two random variables X and Y** G
- [ ] **p(A|B) = P(B|A) for all values of A and B** H

Exam question: Joint Probability Distribution

(3 Points) Consider the following joint probability distribution on random variables A which takes the values 10, 20 and 30 and B which takes the values x, y and z. We will abbreviate p(A=10 & B = z) as p(10, z).

- p(10, x) = 1/6
- p(10, y) = 1/60
- p(10, z) = 1/10
- p(20, x) = 1/60
- p(20, y) = 1/12
- p(20, z) = 1/5
- p(30, x) = 7/20
- p(30, y) = 0/60
- p(30, z) = ?

Which of the following statements are correct (multiple answers possible)?

- [ ] p(30, z) cannot be computed from the information given
- [x] p(x) = 32/60
- [x] **p(A & B) = p(B & A) for all values of A and B**
- [ ] **p(20 | y) = 5/60**
- [x] **p(30, z) = 1/15**
- [x] **p(y | 30) = p(30 | y)**
- [ ] **p(X|Y) = P(Y|X) is never true for all values of two random variables X and Y**
- [ ] **p(A|B) = P(B|A) for all values of A and B**

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Answer the 5
questions

Step 3

Sit back & relax

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Thank you for your attention and participation!

WhatsApp